

Problem Set 3, Question 2(a)

Short-Run Equilibrium Explained

ECO310: Math Intermediate Microeconomics

Spring 2026

1 What is Short-Run Equilibrium?

(Lecture 11-12, slide 31)

From the slides:

“In equilibrium, demand equals supply. Equilibrium price: the price p at which $Q^S(p) = Q^D(p)$ ”

“This is the **ONLY** requirement for a SR equilibrium”

In Plain English:

Short-run equilibrium is when the market clears:

- **Supply = Demand**
- The number of firms N is **fixed** (firms can't enter or exit in the short run)
- Each firm produces whatever maximizes their profit at the market price

2 Problem 2(a) Setup

Given:

- Market demand: $Q^D(p) = 200 - 5p$
- $N = 10$ firms (fixed in short run)
- Each firm has cost: $C(q) = f(N) + \alpha q^2$
 - For part (a): we're just told $N = 10$, and $\alpha > 0$
 - $f(N)$ is a “fixed cost” that depends on how many firms are in the market

Find:

The short-run equilibrium price

3 Solution Steps

(Following Lecture 11-12, slides 17-21, 29-31)

3.1 Step 1: Find each firm's supply curve

Each **price-taking firm** maximizes profit by producing where $p = MC$ (Lecture 11-12, slide 7)

Cost function:

$$C(q) = f(N) + \alpha q^2$$

Marginal cost:

$$MC(q) = \frac{dC}{dq} = 2\alpha q$$

Each firm's supply curve (Lecture 11-12, slide 20):

$$p = MC$$

$$p = 2\alpha q$$

$$q = \frac{p}{2\alpha} \quad \leftarrow \text{This is each firm's supply curve } q^S(p)$$

3.2 Step 2: Find market supply

(Lecture 11-12, slides 29-30)

From the slides: “To derive market supply, we add up the individual market supply curves across firms”

With $N = 10$ firms:

$$Q^S(p) = N \times q^S(p) = 10 \times \frac{p}{2\alpha} = \frac{5p}{\alpha}$$

3.3 Step 3: Set supply = demand

(Lecture 11-12, slide 31)

Equilibrium condition:

$$Q^S(p) = Q^D(p)$$

$$\frac{5p}{\alpha} = 200 - 5p$$

$$\frac{5p}{\alpha} + 5p = 200$$

$$5p \left(\frac{1}{\alpha} + 1 \right) = 200$$

$$5p \left(\frac{1 + \alpha}{\alpha} \right) = 200$$

$$p = \frac{200\alpha}{5(1 + \alpha)}$$

$$p^* = \frac{40\alpha}{1 + \alpha}$$

Final Answer:

$$p^* = \frac{40\alpha}{1 + \alpha}$$

4 Key Differences: Short Run vs. Long Run

Short Run (SR)	Long Run (LR)
Number of firms N is fixed	Firms can enter/exit
Only condition: Supply = Demand	Additional condition: Firms make zero profit
Firms might make profit or loss	$p = \min AC$ (firms break even)

From slide 32: “If firms currently in the market are making money, new firms will come in... if firms are losing money, some will exit”

In the **short run**, we don’t worry about profits yet — we just find where supply meets demand with the current number of firms!

5 Intuition Check

Think of it like a farmers market:

- **Short run:** There are 10 farmers at the market today (can’t change)
- Each farmer decides how much to produce based on the market price
- The price adjusts until the total amount farmers bring = total amount customers want
- Some farmers might make money, some might lose money, but they’re stuck there for today

That’s short-run equilibrium!